

placement of up to four different small boards side-by-side.

With Assembléon's A-series solution too, a manufacturer can simply hire additional ultra-compact robot heads to handle any temporary boosts in production with the same machine.

One can start with 50,000 CPH and go up to 165,000 CPH such as in Assembléon machine. This machine also has the state-of-art 'real-time placement force control' to get the best placement quality.

Quick changeover

Flexibility in SMT production means catering to short-term change of priorities, rush jobs and small production volumes. Several 'costs' are accrued in such an environment—the largest ones being machine setup, setup change and product changeover. In order to keep these costs within control and to enable uninterrupted production, SMT equipment—including placement tools and screen printers—support high-speed changeover.

As far as placement machines go, two changeover strategies are commonly used—one to retrofit the required changes alone and the other to treat the change as a completely new setup to ensure completely optimised position of the feeders. Either way, one must remember that every second a machine is idle, money is lost. But with SIPLACE LES software you can optimise the setup sequence and manage setup changeovers even for the smallest size without stopping your production in most cases.

Therefore machines should ideally support offline setup and programming of the feeders while the machine is still running so that the changeover can be very swift. The trend is also in favour of intelligent feeders to ensure quality.

Small wonder

Smaller the better! That is today's mantra. Remember that Intel ad where an engineer searches for a chip as big as a speck of dust? Well, they did not overdo it. Electronic components are getting smaller and smaller and SMT equipment



Siemens' SIPLACE SX

have to be capable of not just handling these components but also placing them really close together, so that more can be packed into a smaller space!

The size of component packages (resistors, capacitors, etc) is going down further—from 0201 (0.6 mm×0.3 mm) to 01005 (0.4 mm×0.2 mm). SMT equipment must be capable of handling such small component packages, with a mount precision of around 30 microns. Companies like Assembléon also have machines capable of 20 microns, operating with 3-sigma accuracy.

What is more, SMT machines themselves are getting smaller!

Better watchdogs

Quality is built into the SMT production process right from the design stage. Monitoring continues at every stage, be it assembly or soldering or printing. Advanced in-line inspection systems, especially high-speed and high-resolution AOI systems, are the norm today.

State-of-the-art AOI systems include capabilities to inspect 01005 components, high-speed pre- and post-reflow inspection capabilities, advanced vision technologies that learn as the process varies, ability to inspect any component or feature such as chips, ICs, empty locations, lead-free solder

joints, clips and screws. Some recent systems such as the award-winning CyberOptics' Flex HR AOI system include assembly variant programming, charts for process tracking and inspection speed improvements.

Soldering robots

Robots are everywhere—from warfields to factory floors—and the SMT assembly line is one of their favourite playfields. This is especially so during the soldering stage where companies are trying to eliminate human intervention even for selective soldering required in the case of components that cannot be soldered using the reflow process for temperature-sensitive parts. Small, sleek, highly-programmable robots with long arms are becoming very popular in this space. 'Desktop models' are the in-things today.

These robots now come with built-in soldering programs that can be easily handled by even those who do not have professional programming knowledge. Automatic tip-correction, multi-language support, easy parameter setup, etc are other popular features.

Screen printing

Screen printing is a common method for placing solder paste onto PCBs